



WHAT'S NEW IN DOLFINY - HIGH-LEVEL CONVENIENCE WRAPPERS FOR DOLFINx

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Submission type: Poster

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Dolfiny is a collection of convenience wrappers and extended functionality for DOLFINx, first presented in FEniCS 2021 [1] and continuously developed. Over time, the number of features has grown and covers wrappers for PETSc SNES and SLEPc, static condensation, arc-length load control, ODE integration, function space restriction and other minor helpers. More recently we have added support for physical units [2], dimensional analysis [3] and a PETSc TAO wrapper [4]. In this poster contribution we provide an overview of its capabilities and plans for the future.

References

1. Zilian, A., Habera, M. (23 March 2021). *dolfiny: Convenience wrappers for DOLFINx*. FEniCS 2021, Cambridge, United Kingdom.
2. Habera, M., Zilian, A. (18 June 2025). *Physical units in UFL*. FEniCS 2025, Groningen, Netherlands.
3. Zilian, A., Habera, M. (20 June 2025). *Dimensional analysis in UFL*. FEniCS 2025, Groningen, Netherlands.
4. Kühner, P. T., Habera, M., Zilian, A. (18 June 2025). *PETSc TAO support for optimisation problems with FEniCSx*. FEniCS 2025, Groningen, Netherlands.